

Move Over, Jetsons

Prefab, manufactured, modular homes. What do these words make you think of? Ultramodern Buckminster Fuller spaceship like dwelling solutions for ultramodern people, and shiny, hard surfaces—a far cry from living in cooperation with nature. Trailer parks even come to mind when I think of modular homes, complete with plastic flamingos and rainbow whirligigs. The first manufactured homes came out in the Gold Rush era in California, when many buildings were needed in a hurry and these were constructed on-site from kits. They provided quick and dirty housing, and architecture was not an issue. Today, “prefab” is gaining a new reputation with not only cutting-edge architecture, but environmentally sustainable designs, from manufacture to end use.

The financial upsides of prefab dwellings over site-built are many. The materials savings in the factory are far better for factory-built homes than for site-built. Whereas cutoffs on-site end up in the job site dumpster, cutoffs in the factory can be reused for the next project, and prefab houses can be designed to limit this waste. Since factory-built houses take less time to assemble, less of the land’s mortgage price is spent on the owners not living there. There are fewer insurance costs while builders work. Weather is not an issue when homes are mostly built inside a factory. Workers in the factory environment can become more specialized at their repetitive roles under better supervision. Buildings can be placed in difficult-to-build sites, in pieces for assembly. The headaches saved by having a simpler, predesigned plan for the customer to work with are a significant plus.

Two California prefab home builders, LivingHomes and Michelle Kaufmann Designs, are leading the way on the West Coast with well-designed



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Features of Prefab Homes

Features of these green prefab homes include, but are not limited to,

- FSC-certified woods;
- low-flow plumbing fixtures;
- natural lighting;
- durable, travel-worthy framing;
- Energy Star appliances;
- living roofs;
- graywater and water catchment systems;
- low- and no-VOC paints;
- passive-solar siting;
- foam insulation;
- permeable driveways;
- recycled-material countertops;
- LED and CFL lighting systems and controls; and
- radiant on-demand water heaters.

dwellings for real people (see “Features of Prefab Homes”). Both builders’ designs also happen to be built using state-of-the-art sustainable features, from their FSC-certified flooring to their Energy Star appliances, making the homes sustainable to build as well as energy efficient to dwell in. LivingHomes, in Southern California, is a maker of architect-designed, high-end prefab and modular homes, whose emphasis is on green prefab. Their model home earned the first LEED Platinum certification for a residential building (see “Prefab Home Earns LEED Platinum,” *HE*

Jan/Feb ’07, p. 10). Floor plans range from 650 to 6,000 square feet. While it’s difficult to predict the actual cost of a house, since permitting, foundations, land, and transportation will all have costs unique to the building site, costs—minus taxes and the other fees mentioned—can range from \$150 to \$250 per square foot. LivingHomes architects Ray Kappe and David Hertz both have strong environmental backgrounds, and their designs celebrate the indoor-outdoor beauty of California living. “I’ve always sought out the edges, the views, and a feeling

of expansiveness,” Kappe says of his designs.

Michelle Kaufmann’s homes are becoming a household word in the sustainable-building communities in the San Francisco Bay Area. With an elaborate installation during the 2007 West Coast Green Conference, including landscaping, the mkLotus House was visited by thousands of people over the weekend. Like LivingHomes designs, Kaufmann’s designs emphasize bringing the outside in, with accordion like wall panels of glass doors, passive-solar siting, and trendy interiors, all made from sustainable products. After moving in, the homeowner can enjoy energy savings from the Energy Star appliances, natural light, and cross-ventilation. Buildings are also wired for future installation of a PV system, to name just a few features. Costs for a Kaufmann-designed manufactured home are around \$250 per square foot,

the average for any type of home in the San Francisco Bay Area. One of my favorite features of Kaufmann’s designs is the emphasis on small. As she says, “It is not necessary to build a big house, but rather to design one that feels spacious. By opening this house to the outdoors, it feels much larger than it is, and ‘borrows’ the sense of space from the landscape.”

While these two makers are focusing on green materials and methods in their designs, the real proof of their sustainability will be in their end users’ experiences, and the utility bills down the road. A home is only as green as its residents, after all. Can these homes be easily inserted into an urban setting as infill? Can an existing building be remodeled using prefab, mass-produced parts? Can a prefab building be remodeled? What are the reuse potentials of the homes’ parts after the building has finished being a home? One of the

green building

building features overlooked in many building styles is the waste potential of a building after its use. With the continued evolution and resurgence of this kind of building, perhaps we will see answers to these questions. With their modern, stylish, and accessible angle on green, Kaufmann Designs and LivingHomes are changing the way we think about prefab, and raising the bar for green building in general.

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—Leslie Jackson

*Leslie Jackson is **Home Energy**’s new associate editor.*

For more information:

For more on LivingHomes, go to www.livinghomes.us.

For more on Michelle Kaufmann Designs, go to www.mkd-arc.com.